TIME AND WORK

1). If A takes x days to finish the work, his one day work is $\frac{1}{x}$.

$$\boxed{1 \text{ day work} = \frac{1}{x}}$$

2. Efficiency is work done. so efficiency or work done is inversely proportional to no. of days / hours. $\boxed{E (or) W \propto \frac{1}{\text{No. of days}}}$

3. Money is distributed based on efficiency ratio, so money for work is directly proportion to efficiency

$$\boxed{\text{Money} \propto \text{Efficiency}}$$

4. If A takes x days to finish the work, and B takes y days to finish the work, then working together they finish $\frac{xy}{x+y}$ days.

$$\left.\begin{array}{l} A \rightarrow x \text{ days} \\ B \rightarrow y \text{ days} \end{array}\right\} A+B = \frac{xy}{x+y} \text{ days}$$

5. Total work is always 100% ie Total work = 100% = 1

6. Work remaining $\boxed{= 1 - \text{work completed}}$

1) A can do 75% of a work in 9 days and B can do half of the same work in 8 days. In how many days they will finish the work, working together.

$$\boxed{\text{Ans: } 6\frac{6}{7} \text{ days}}$$

75% 9 days

half 8 days

$$\frac{3}{4} \xrightarrow{A} 9 \text{ days}$$

$$1 \longrightarrow 9 \times \frac{4}{3} = 12 \text{ days}$$

$$\frac{1}{2} \xrightarrow{B} 8 \text{ days}$$

$$1 \longrightarrow 8 \times \frac{2}{1} = 16 \text{ days}$$

$$A+B \rightarrow \frac{12 \times 16}{12+16} = \frac{48}{7}$$

$$= 6\frac{6}{7} \text{ days}$$

2. A can do $\frac{2}{5}$th of the work in 12 days while B can do $62\frac{2}{3}$% of work in 16 days. They work together for 10 days. In how many days B alone can complete the remaining work.

$\boxed{\text{Ans: 6 days}}$

$\frac{2}{5} \xrightarrow{A} 12$ days $\quad 1 \xrightarrow{A} 12 \times \frac{5}{2} = 30$ days

$\frac{2}{3} \xrightarrow{B} 16$ days $\quad 1 \xrightarrow{B} 16 \times \frac{3}{2} = 24$ days

For 10 days,

$\Rightarrow 10\left(\frac{1}{30} + \frac{1}{24}\right)$

$\Rightarrow 10\left(\frac{24+30}{30 \times 24}\right)$

$\Rightarrow \frac{3}{4} \Rightarrow$ work completed

Remaining $\Rightarrow 1 - \frac{3}{4} = \frac{1}{4}$ (remaining)

$1 \xrightarrow{B} 24$ days

$\frac{1}{4} \longrightarrow 24 \times \frac{1}{4} = 6$ days

3. To do certain work, the efficiency ratio of A and B is 3:7. Working together they can finish the work in 10.5 days. They worked together for 8 days. In how many days A alone can complete 60% of the remaining work.

$\boxed{\text{Ans: 5 days}}$

Days $\Rightarrow$ 7 : 3

$A \rightarrow 7x = 35$ days

$B \rightarrow 3x = 15$ days

$\frac{7x \times 3x}{10x} = \frac{21}{2}$

$\boxed{x = 5}$

$\Rightarrow 8\left[\frac{1}{35}+\frac{1}{15}\right] = 8\left[\frac{50}{35\times 15}\right] = \frac{16}{21}$

Remaining $\Rightarrow 1-\frac{16}{21} = \frac{5}{21}$

$1 \xrightarrow{A} 35$ days

60% of $\frac{5}{21} \Rightarrow \frac{\cancel{60}^{3\,1}}{\cancel{100}_{20}} \times \frac{\cancel{5}^{1}}{\cancel{21}_{7}} = \frac{1}{7}$

$\frac{1}{7} \xrightarrow{A} \frac{1}{7}\times 35 = 5$ days

4). A takes 10 days less than B to finish a piece of work If both A and B together can do a work in 12 days then in how many day B alone can finish complete work.

$\boxed{\text{Ans: 30 days}}$

$B \rightarrow x$ days

$A \rightarrow (x-10)$ days

$\frac{x(x-10)}{x+x-10} = 12$

$A \rightarrow 30 = (x-10)$ days.

$x^2-10x = 24x-120$

$x^2-30x-4x+120=0$

$x(x-30)-4(x-30)=0$

$(x-30)(x-4)=0$

↳ because $x=4$ sub

$(x-10) \Rightarrow 4-10=-6$

Not valid

$\boxed{\text{So. } x=30}$

5). A can do half of the work in 5 days, B can do 3/5 th of same work in 9 days, C can do 2/3 rd of same work in 8 days working together they can do their finish in how many days?

$\boxed{\text{ANS: 4 days}}$

$A \rightarrow \frac{1}{2} \rightarrow 8$ days $\Rightarrow A \rightarrow 10$ days

$B \rightarrow \frac{3}{5} \rightarrow 9$ days $\Rightarrow B \rightarrow 15$ days

$C \rightarrow \frac{2}{3} \rightarrow 8$ days $\Rightarrow C \rightarrow 12$ days.

$$A+B+C \xrightarrow{1\text{ day}} \frac{1}{10}+\frac{1}{15}+\frac{1}{12} = \frac{12+8+10}{120} = \frac{1}{4}.$$

$A+B+C \rightarrow 4$ days

6) A is thrice as good work men as B. C alone takes 48 days to finish a work. working together all the three can finish the work in 16 days. In how many days B alone can finish the complete work.

$\boxed{\text{Ans: 96 days}}$

Efficiency $A:B = 3:1$

No. of days $A:B = 1:3$

$A \rightarrow x$

$B \rightarrow 3x \rightarrow 3\times 32 = 96$ days

$C \rightarrow 48$

$A+B+C \rightarrow 16$ days

$A+B+C \rightarrow \frac{1}{16}$

$$\frac{1}{x}+\frac{1}{3x}+\frac{1}{48} = \frac{1}{16}$$

$$\frac{3x+x}{3x^2}+\frac{1}{48} = \frac{1}{16}$$

$$\frac{4x}{3x^2}+\frac{1}{48} = \frac{1}{16}$$

$$\frac{4}{3x}+\frac{1}{48} = \frac{1}{16}$$

$$\frac{4\times 48+3x}{48\times 3x} = \frac{1}{16}.$$

$$184+3x = \frac{\cancel{48}^{\cancel{24}\ \cancel{12}\ 3}\times 3x}{\cancel{16}\ \cancel{8}\ \cancel{4}}$$

$184+3x = 9x$

$6x = 184$

$\boxed{x = 32}$

7. A can do 40% of work in 12 days, B can do 60% of the same work in 15 days Both worked together for 10 days. And C completes the remaining work alone in 4 days. working together all the three can complete 28% of the same work in how many days?

$\boxed{\text{Ans: 2 days}}$

$A \rightarrow 40\% \rightarrow 12$ days

$B \rightarrow 60\% \rightarrow 15$ days

$\frac{2}{5} \xrightarrow{\cancel{A}} 12 \Rightarrow 1 \xrightarrow{A} 30$ days

$\frac{3}{5} \xrightarrow{B} 15 \Rightarrow 1 \xrightarrow{B} 25$ days

$A+B+C \overset{1\text{ day}}{\Longrightarrow} \frac{1}{30} + \frac{1}{25} + \frac{1}{15}$

$\Rightarrow \frac{10+12+20}{300}$

$\Rightarrow \frac{42}{300}$

Together,

$10\left[\frac{1}{30} + \frac{1}{25}\right]$

$\Rightarrow 10 \times \frac{55}{30 \times 25} \Rightarrow \frac{11}{15}$

$\frac{4}{15} \xrightarrow{C} 4$ days

(Remaining)

$1 \xrightarrow{C} 15$ days

$A+B+C \cdot 100\% \longrightarrow \frac{300}{42}$ days

$28\% \longrightarrow x$

$\Rightarrow \frac{\cancel{300}^{1}}{\cancel{42}_{\cancel{14}_{1}} \times \cancel{100}} \times \cancel{28}^{2}$

$\Rightarrow 2$ days.

Model - 2.

8). A & B can do a piece of work in 72 days. B and C can do in 120 days, while A & C can do it in 90 days. working together in how many days they can complete the same work?

$\boxed{\text{Ans: 60 days}}$

$A+B \rightarrow \frac{1}{72}$

$B+C \rightarrow \frac{1}{120}$

$C+A \rightarrow \frac{1}{90}$

$2(A+B+C) = \frac{1}{72} + \frac{1}{120} + \frac{1}{90} = \frac{10+6+8}{720} = \frac{1}{30}$.

$A+B+C = \frac{1}{30 \times 2} = \frac{1}{60}$

So, $A+B+C \Rightarrow 60$ days

a). A & B can do a piece of work in 8 days, B and C do it in 24 days, while C & A can do it in $8\frac{4}{7}$ days. In how many days C alone can finish the same work.

$\boxed{\text{Ans: 60 days}}$

$A+B \rightarrow \left(\frac{1}{8}\right)$ or 8 days

$B+C \rightarrow \frac{1}{24}$ or 24 days

$C+A \rightarrow \frac{7}{60}$ (or) $\frac{60}{7}$ days

$2(A+B+C) = \frac{1}{8} + \frac{1}{24} + \frac{7}{60} = \frac{34}{120}$

$A+B+C = \frac{34}{240}$.

$\therefore C = (A+B+C) - (A+B)$

$= \frac{34}{240} - \frac{1}{8} \times \frac{30}{30}$

$= \frac{34}{240} - \frac{30}{240} = \frac{4}{240}$

$= \frac{1}{60}$.

$\therefore$ C alone $\longrightarrow$ 60 days.

10. A does 2/5 th of the work in 9 days. Then B joined him and together they completed the remaining work in 6 day. In how many days, B alone can finish the work?

$\boxed{\text{Ans: 18 days}}$

$\frac{2}{5} \xrightarrow{A} 9$ days

$1 \longrightarrow \frac{45}{2}$ days

Remaining,

$\frac{3}{5}(A+B) \longrightarrow 6$

$A+B \longrightarrow 10$ days.

$A+B \longrightarrow \frac{1}{10}$

$B = \frac{1}{10} - \frac{2}{45}$

$\boxed{B = \frac{1}{18}}$

$\therefore B = 18$

11). If A, B, C can finish a work in 12, 18, 36 days respectively. And they worked together for 2 days. B quits the work. In how many day A & C together can finish the remaining work?

$\boxed{\text{Ans: 6 days}}$

$2(A+B+C) = 2\left(\frac{1}{12} + \frac{1}{18} + \frac{1}{36}\right)$

$= \frac{1}{3}$ work completed.

x days of $(A+C) = \frac{2}{3}$

$x(A+C) = \frac{2}{3}$

$x\left(\frac{1}{12} + \frac{1}{36}\right) = \frac{2}{3}$

$x\left(\frac{\cancel{48}}{\cancel{12} \times \cancel{36}}\right) = \frac{2}{3}$

$x = \frac{2}{3} \times 9$

$x = \frac{\cancel{18}^{6}}{\cancel{3}}$

$\boxed{x = 6}$

12) A & B can do a piece of work in 45 and 40 days respectively. They Began the work together, but A left after some days, B finished the remaining work in 23 days. After how many days A left the work.

$\boxed{\text{Ans: 9 days}}$

Let A left after x days

x days work (A+B) + 23 days work of B = 1

$$x\left(\frac{1}{45}+\frac{1}{40}\right)+23\times\frac{1}{40}=1$$

$$x\left(\frac{40+45}{40\times45}\right)=1-\frac{23}{40}$$

$$\frac{x}{9}=1 \Rightarrow x=9$$

13). A started a work and left after working for 9 day, B finished the remaing work in 24 days Had A left the work after working for 13 days, B would have finished the remaining work in 18 days. In how many A & B working together can finish the work?

$\boxed{\text{Ans: 15 days}}$

A	9 days	24 days
	13 days	18 days

$$\rightarrow \frac{9}{x}+\frac{24}{y}=1\times3$$

$$\rightarrow \frac{13}{x}+\frac{18}{y}=1\times4.$$

A → x days

B → y days

$$\frac{27}{x}+\frac{72}{y}=3$$

$$\frac{52}{x}+\frac{72}{y}=4 \Rightarrow \boxed{x=25}$$

10.

$\Rightarrow \frac{9}{25} + \frac{24}{y} = 1 \quad \Rightarrow \quad y = \frac{75}{2}.$

a

l $\quad x = 25 \quad y = \frac{75}{2} \quad A+B \longrightarrow \dfrac{25 \times \frac{75}{2}}{25 + \frac{75}{2}} = \dfrac{\frac{25\times75}{\cancel{2}}}{\frac{50+75}{\cancel{2}}} = \dfrac{25\times75}{125}$

it

$\boxed{= 15 \text{ days}}$

14) A can do a piece of work in 10 days and B can do it in 12 days. They worked together for 3 days. The B leaves and A alone continues the work. 2 days after C joins the work and completes it in 2 days along with A. In how many days C alone can finish the work.

$\boxed{\text{Ans: 40 days}}$

11)

$3(A+B) = 3\left(\frac{1}{10} + \frac{1}{12}\right) = 3 \times \left(\frac{12+10}{10\times12}\right) = \frac{11}{20}$ work.

or

$2A = 2\times\frac{1}{10} = \frac{1}{5} = \frac{4}{20}.$

it

Remaining,

$\Rightarrow 1 - \left[\frac{11}{20} + \frac{4}{20}\right]$

$\Rightarrow 1 - \frac{15}{20} \Rightarrow \frac{5}{20} = \frac{1}{4}$ work left.

it

$\frac{1}{4}(A+C) \longrightarrow 2$ days

$A+C \longrightarrow 8$ day

$C \longrightarrow \frac{1}{8} - \frac{1}{10}$

$C \longrightarrow \frac{1}{40}.$

$C \rightarrow 40$ days.

15. A & B can finish a work in 20 and 12 days A starts the work alone & after 4 days B joined him till the completion of the work. How long did the work last ?

$\boxed{\text{Ans: 10 days}}$

$A(A) = 4 \times \frac{1}{20} = \frac{1}{5}$ work.

Total no. of day

$= 4 + 6 = 10$ days

$\frac{4}{5} = x(A+B)$

$x = \frac{4}{5(A+B)}$

$x = \frac{4}{5\left(\frac{1}{20} + \frac{1}{12}\right)}$

$x = \frac{4}{\frac{1}{4} + \frac{5}{12}} = \frac{4}{\frac{12 \times 4}{48}} = \frac{4 \times 48}{12 \times 4} = 6$

$\boxed{x = 6}$

16. A can do a certain work in 12 days. B is 60% more efficient than A. In how many days A & B working together can finish the same work?

$\boxed{\text{Ans: } \frac{60}{13}}$

~~$E \propto \frac{1}{D}$~~

$\boxed{\frac{E_A}{E_B} = \frac{D_B}{D_A}}$

$\frac{100}{160} = \frac{D_B}{12}$

$\frac{12 \times 15}{39} = \frac{60}{13}$

$D_B = \frac{15}{2}$ days

$D_A = 12$ days

$A+B \rightarrow \frac{12 \times \frac{15}{2}}{12 + \frac{15}{2}}$

~~$\frac{\frac{12 \times 15}{2}}{\frac{24+15}{2}}$~~

Ans:- $E \propto 1/D$

$E_A / E_B = D_B / D_A$

$100/160 = D_B/12$

$D_B = \frac{15}{2}$ day

$D_A = 12$ day.

$A+B : \frac{12 \times 15/2}{12 + 15/2} = \frac{12 \times 15/2}{39/2}$

$= \frac{12 \times 15}{39}$

$= \frac{60}{13}$

17. The efficiency of A, B & C are in the ratio 5:3:2 working together, they can complete a task in 21 hrs. In how many hrs, B alone can complete 40% of the task?

$\boxed{\text{Ans: 28 hrs}}$

5 : 3 : 2

$A \Rightarrow \frac{10}{5} \times 21 = 42$ hrs

$B \Rightarrow \frac{10}{3} \times 21 = 70$ hrs

$C \Rightarrow \frac{10}{2} \times 21 = 105$ hrs

$100\% \xrightarrow{B} 70\%$

$40\% \xrightarrow{B} x$

$\Rightarrow \frac{70}{100} \times 40$

$\Rightarrow 28$ hrs.

18. To do a certain work, the ratio of efficiency of A, B, C is 7:5:6. Working together, they can complete same work in 35 days. B & C together worked for 21 days, in how many days, A alone can complete the remaining work?

$\boxed{\text{Ans: 57 days}}$

$A \longrightarrow \frac{18}{7} \times 35 = 90$ days

$B \longrightarrow \frac{18}{5} \times 35 = 126$ days

$C \longrightarrow \frac{18}{6} \times 35 = 105$ days

$21(B+C) = 21\left(\frac{1}{126} + \frac{1}{105}\right)$

$= \frac{11}{30}$

Remaining work $= \frac{19}{30}$

$1 \xrightarrow{A} 90$ days

$\frac{19}{30} \xrightarrow{A} ?$

$\frac{19}{30} \times 90 = 57$ days

19. If A & B can do a piece of work in 9 & 15 day. If they work for a day alternatly with A begining, then in how many days, work will be completed?

$\boxed{\text{Ans: 11 days}}$

2 days $\xrightarrow{A+B}$ $\frac{1}{9} + \frac{1}{15} = \frac{8}{45}$

$\downarrow \times 5$ $\qquad\qquad$ $\downarrow \times 5$ $\qquad$ to make close to 1 not greater than 1

10 days $\longrightarrow$ $\frac{40}{45}$

10 days $\rightarrow$ $\frac{40}{45} = \frac{8}{9}$

A $\xrightarrow{1 day}$ $\frac{1}{9}$ $\rightarrow$ 1 day. $\rightarrow$ remaining work also 1/9 so, with 1 extra day A completes.

Total No of day : 10+1 = $\boxed{\text{11 day}}$

20. A is 1.5 times efficient than B and therefore takes 8 days less than B to finish a work. If A & B works on alternate day with A begining, in how many days work will be completed.

$\boxed{\text{Ans: 19 days}}$

Efficiency

$A = 1.5B$

$A = \frac{3}{2}B$

$\frac{A}{B} = \frac{3}{2}$

Days $\rightarrow$ 2 : 3

A = 16 days $\qquad$ B = 24 days

A $\longrightarrow \frac{1}{16} \longrightarrow$ 1 day

A+B $\rightarrow \frac{1}{16} + \frac{1}{24}$

2 days

= 5/48

$\downarrow \times 9$ $\qquad$ $\downarrow \times 9$

18 days $\qquad$ 45/48

Total No. of days = 19

(18 + 1) A's extra 1 day work

Remaining $\left[\frac{3}{46} = \frac{1}{16}\right]$.

21. A, B, C can finish a work in 30, 20, 10 days. A is assisted by B on 1 day and C on the next day alternately. In how many days the work will be completed?

$\boxed{\text{Ans: } 9\frac{3}{8} \text{ days}}$

$2 \text{ days} \xrightarrow{2A+B+C} \frac{2}{30} + \frac{1}{20} + \frac{1}{10}$

$= \frac{4+3+6}{60} = \frac{13}{60}$

×4 ×4

8/60 → Remaining

$8 \text{ days} \longrightarrow 52/60$.

[Total 8+1=9].

$A+B \xrightarrow{9^{th} day} \frac{1}{30} + \frac{1}{20} = \frac{1}{12} = \frac{5}{60}$.

$A+C \xrightarrow{10th\ day} \frac{1}{30} + \frac{1}{10} = \frac{40}{30 \times 10} = \frac{4}{30} = \frac{8}{60}$.

$\frac{3}{60}$ $\quad \frac{3}{8}$ day $\quad 9 + \frac{3}{8}$ day.

22. If 3 men or 4 women can do a piece of work in 43 days. Then in how many days 7 men & 5 women takes to finish the same work?

$\boxed{\text{Ans: } 12 \text{ days}}$

$3\text{men} \xrightarrow{1day} \frac{1}{43}$ $\quad 1m \xrightarrow{1day} \frac{1}{43 \times 3}$ $\quad 7m + 5w \xrightarrow{1day} 7 \times \frac{1}{43 \times 3} + 5 \times \frac{1}{43 \times 4}$

$4w \xrightarrow{1day} \frac{1}{43}$ $\quad 1w \xrightarrow{1day} \frac{1}{43 \times 4}$.

$= \frac{1}{43} \times \frac{43}{12} = \frac{1}{12}$.

So, 12 days.

23. If 1 men or 2 women or 3 boys can finish a work in ~~44~~ days. Then In how many days 1 men, 1 women, 1 boy finish the same work?

$\boxed{\text{Ans: 24 days}}$

$1m \rightarrow \frac{1}{44}$ $\quad$ $1w \rightarrow \frac{1}{88}$ $\quad$ $1b \rightarrow \frac{1}{44 \times 3}$

$1m + 1w + 1B \rightarrow \frac{1}{44} + \frac{1}{88} + \frac{1}{44 \times 3}$

$= \frac{1}{44}\left[1 + \frac{1}{2} + \frac{1}{3}\right] = \frac{1}{44}\left[\frac{3}{2} + \frac{1}{3}\right] = \frac{1}{\cancel{44}_{4}}\left[\frac{\cancel{11}}{\cancel{6}_{3}}\right] = \frac{\cancel{5}}{\cancel{2}}$

$\boxed{= \frac{1}{24}}$

24. A & B can complete a piece of work in 15 days & 10 days. They contracted to complete the work for Rs. 30000. Find the share of A in the contracted money.

$\boxed{\text{Ans: 12000}}$

Days ratio $A : B \Rightarrow 15 : 10 \Rightarrow 3 : 2$

Efficiency ratio $A : B \Rightarrow 2 : 3$

$5P \rightarrow 30000$

A means 2 parts

$1P \rightarrow 6000$

$2(6000)$

$\Rightarrow 12000.$

25) Two men undertake a job for Rs 960. They can complete the job in 16 days ~~&~~ 24 days when working alone. They work along with 3rd man & ~~took~~ 8 days to finish the work. Find the share of 3rd man in the total money.

$\boxed{\text{Ans: } 160/-}$

$A \rightarrow 16$ days

$B \rightarrow 24$ days

$C \rightarrow x$ days

$A+B+C = \frac{1}{8}$

$\frac{1}{16} + \frac{1}{24} + \frac{1}{x} = \frac{1}{8}$

$x = \frac{1}{48}$ so, $\boxed{x = 48}$

Days ratio $= 16 : 24 : 48$

$= 2 : 3 : 6.$

Efficiency ratio $\left[\frac{1}{2} : \frac{1}{3} : \frac{1}{6}\right] \times 6$

$\Rightarrow 3 : 2 : 1.$

c's share $= \frac{1}{6} \times 960 = 160/-$

26) A & B took a work for Rs. 880. A got Rs. 240 more than B when they worked together, and B will take 12 days more than A, when they work individually. In how many day A & B working together can finish the whole work?

$\boxed{\text{Ans: } 10\frac{2}{11}}$

<u>wages:</u>

$A \rightarrow 240 + x$

$B \rightarrow x.$

$240 + x + x = 880$

$2x = 640$

$\boxed{x = 320}$

Efficiency Ratio = 560 : 420 = 7 : 4

$3p \rightarrow 12$

$1p \rightarrow 4$

Days ratio = 4 : 7

16 days   28 days

$$A + B = \frac{16 \times 28}{44} = \frac{112}{11} = \frac{110}{11} + \frac{2}{11} = 10\frac{2}{11}.$$